Japan:

The state of High Temperature Superconductivity in 2025

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Energy Context

Japan’s final energy consumption in fiscal year (FY) 2024 totaled approximately 11.31 exajoules, continuing its downward trend since the mid-2000s. This is primarily attributable to sustained improvements in energy efficiency in the industrial and residential sectors. Electricity accounted for approximately 30% of final energy consumption. The share of non-fossil power sources had declined following the Great East Japan Earthquake in 2011, but it has been growing again in recent years, ultimately reaching 19.9% of primary energy supply. This reflects the expanded deployment of renewable energy as well as the partial restart of nuclear power plants. In terms of the electricity generation mix, the combined share of non-fossil power sources (specifically, renewable energy and nuclear power) exceeded 30%. This indicates steady progress in the ongoing transition of Japan’s power generation mix. By contrast, Japan’s energy self-sufficiency rate remains low at 16.4%, and a high level of dependence on imported fossil fuels continues to represent a significant structural challenge.

Policy, Regulatory, and Infrastructure

Japan’s energy policy is fundamentally oriented toward accelerating the maximum deployment of renewable energy while ensuring a stable supply of electricity over the medium to long term. To achieve decarbonization, economic growth, and energy security simultaneously, the government is advancing institutional and regulatory reforms in a phased manner. This approach emphasizes taking comprehensive stock of the power generation mix, investment environment, and supply-demand balancing mechanisms.

Institutional Reforms and Measures to Promote Renewable Energy

Japan has set a target for renewable energy to account for 40 to 50% of the power generation mix by FY2040. In line with this target, the government is advancing a transition from a feed-in tariff to a feed-in premium scheme, shifting toward a market-integrated support mechanism. In anticipation of the large-scale deployment of renewable energy, priority is being placed on investments in transmission grid reinforcement and the development of interregional interconnection lines. At the same time, institutional use of battery storage systems and demand response are expanding.

Climate Change and GX (Green Transformation) Policies

Japan has set a goal for FY2040 of reducing greenhouse gas emissions by 73% compared to FY2013 levels, and also aims to achieve carbon neutrality by 2050. Policies are being implemented to strengthen energy efficiency and promote electrification, and the development of long-term investment support frameworks for renewable energy, hydrogen and ammonia, and carbon capture and storage are progressing with an eye toward the 2040 target.

Energy Supply Security and Nuclear Energy Policy

Electricity demand is expected to grow over the medium to long term due to the increasing number of data centers and semiconductor manufacturing facilities. In light of these developments, the government clarified its nuclear power policy as one of continued usage with safety as the top priority. Specifically, the policy calls for nuclear power to make up an approximate 20% share of the electricity generation mix by FY2040, with institutional arrangements and operational measures being advanced that emphasize safety.

R&D Activities

NEDO Leading Research Program/Frontier Development Program

The New Energy and Industrial Technology Development Organization (NEDO) launched the Frontier Development Program to promote research and development and subsequent commercialization in frontier areas that Japan should address nationally, with the aim of creating new industries beyond 2040. In FY2025, “Extreme Materials” was designated as a target area, and integrated initiatives are being promoted that span the development of a technology through to its real-world social implementation.

Technology That Extends the Maximum Length of HTS Spiral Assembled Conductors

This research aims to overcome manufacturing constraints arising from the limited maximum length of individual high-temperature superconducting (HTS) tapes. Overcoming this critical bottleneck would allow the realization of extended-length spiral assembled conductors applicable to next-generation superconducting power equipment, including large-scale power apparatuses and high-field electromagnets. The study is based on the premise of employing butt joint technology for superconducting tapes rather than conventional ultra-low-resistance joints, and focuses on optimizing cooling stability at joint sections. The mechanical properties and current-sharing characteristics of the assembled conductors are also being studied. In addition, the project aims to establish a technical foundation for extending conductor length beyond the limitations of individual HTS tapes by establishing real-time, non-contact evaluation technologies to assess the structural integrity of HTS spiral assembled conductors during winding processes.

Industrial Rare Earth-Based High-Temperature Superconductors

This research focuses on the development of industrial rare earth-based HTS conductors (REBCO conductors) suitable for large-scale, meter-class, high-field industrial equipment. The objective is to realize REBCO conductors that exhibit low alternating-current (AC) losses while simultaneously demonstrating superior industrial-grade mechanical properties, including high strength. The study investigates a tape fabrication process based on an original grain-boundary control method and conducts structural analyses in parallel with conductor fabrication to develop REBCO conductors with enhanced AC loss performance. Additionally, by advancing analyses of the electromagnetic response of the conductors, the project seeks to theoretically elucidate the mechanisms responsible for AC loss reduction in industrial REBCO conductors.

Research and Development of Assembled High-Temperature Superconductors for Advanced-Performance Magnets

This research involves the development of assembled HTSs for advanced-performance magnets based on HTS tapes with a multifilament structure known as an inter-filament bridge, fabricated using high-speed, high-precision laser processing technologies. The design of these spiral copper-plated striated coated-conductor cables with inter-filament bridges (SCSC-IFB cables) and the establishment of manufacturing technologies suitable for mass production can effectively suppress the degradation of current-carrying performance in assembled HTSs. This approach simultaneously enables reduced AC losses as well as the high strength and flexibility required for advanced-performance magnetics. The assembled HTSs that are developed can be employed in the fabrication of complex-shaped coils for use in applications ranging from medical devices to particle accelerators, electric energy storage systems, and fusion.

Program to Develop an Innovative Propulsion System for Aircraft

This program aims to be the first in the world to conduct research, development, and demonstration activities to apply superconducting technologies—key to electric aircraft propulsion—to next-generation aircraft. As a way to make strides toward the achievement of carbon neutrality by 2050, it focuses on applying electric propulsion to large aircraft, an area that faces challenges due to weight and power output constraints. To address these challenges, the program is developing superconducting systems, including superconducting generators, motors, cables, and inverters, that enable high efficiency and high power output.

Development of 2 MW Class System Technologies

This initiative involves the development of lightweight superconducting generators suitable for onboard aircraft installation, fully superconducting AC motors, and lightweight cooling systems. The ultimate aim is to construct a 2 MW class superconducting system for ground-based testing.

Development of Advanced System Element Technologies

Element technologies and system specifications are being developed to enable the realization of a superconducting system with output scaled up to approximately 20 MW, the amount required for propulsion of wide-body aircraft. These development efforts are being advanced with the objective of linking the results to flight validation of next-generation aircraft, particularly small- and medium-sized aircraft.

Formulation of Plans for Real-World Social Implementation

To prepare for the installation of superconducting systems in aircraft, the systems will be standardized and plans are being made to gain safety and performance certification. Surveys of standardization trends for electric aircraft will be conducted and issues associated with incorporating superconducting specifications into international standards will be identified.

JST - Mirai Program: High-Temperature Superconducting Wire Joint Technologies Leading to Innovative Reduction of Energy Loss

Superconductivity technologies are already used in applications such as nuclear magnetic resonance, magnetic resonance imaging, and superconducting magnetic levitation railways. However, low-temperature superconducting materials that require cooling with liquid helium are the current dominant technology. This expensive cooling represents a major obstacle to the widespread adoption of superconductivity technologies.

HTS materials are considered more suitable for full-scale, real-world social implementation, as they can be cooled using relatively low-cost liquid nitrogen and are capable of generating much higher magnetic fields than low-temperature superconductors. However, HTSs can typically only be manufactured in lengths of around 500 hundred meters. Therefore, it is essential to establish reliable joining technologies that can maintain superconductivity and provide extremely low electrical resistance.

This project focuses on the challenging development of HTS conductor joint technologies, specifically, the following research, implementation, and demonstration activities:

- Implementation and demonstration of a 1.3 GHz, 30.5 T NMR magnet—the world's highest-field magnet—using superconducting joints (10^{-13} Ω)
- Implementation of liquid-nitrogen-cooled superconducting railway feeder cables using ultra-low-resistance joints (10^{-7}–10^{-8} Ω)

Through this project, high-efficiency, high-magnetic-field coils and long-distance direct-current power transmission will be brought significantly closer to realization, accelerating the full-scale, real-world social implementation of superconductivity technologies. Source: Japan Science and Technology Agency (JST), ALCA-Next Program

JST ALCA-Next: Low AC Loss and Robust High-Temperature Superconductor Technology

HTSs allow for high current density with no electrical resistance, and therefore offer significant potential for energy savings as well as the miniaturization and weight reduction of electronics. However, their practical application has been hindered by several challenges: they are subject to AC loss; achieving high current capacity is difficult; and the tape-shaped geometry of coated conductors limits their strength and flexibility.

A multifilament (striated) structure, with a thin superconducting film divided into narrow filaments, is effective in reducing AC loss, but these striated coated conductors are vulnerable to local defects and normal transition. Robustness can be improved by copper plating of the entire thin film to allow current bypassing, but this approach results in additional AC loss due to coupling currents.

An SCSC cable has been developed in an attempt to overcome these challenges. These feature copper-plated, multifilament coated conductors spiral-wound in multiple layers on a round metal core. This cable structure reduces AC loss and improves robustness against normal transition while maintaining multidirectional bendability. In the feasibility study phase, this approach successfully demonstrated an approximately one-tenth reduction in AC loss compared to standard coated conductors and 1 kA-class AC current transport. In the full-scale R&D phase, the project aims to produce demonstrator coils that advance the technology toward ~2 kA-class current capacity, higher-frequency operation, and applicability to coils with various three-dimensional shapes.

These developments should allow the SCSC cable technology to be utilized in R&D for advanced electrical devices that contribute to carbon neutrality, such as superconducting motors for aircraft propulsion and superconducting magnetic energy storage systems (SMES) that support the introduction of renewable energy into power grids. Source: Japan Science and Technology Agency (JST), ALCA-Next Program

Next Term

Japan has entered a phase in which it pursues renewable energy expansion and advances electrification while simultaneously ensuring the security of its energy supply. To achieve its 2030 targets and ultimately carbon neutrality by 2050, the government is promoting the deployment of renewables and non-fossil power sources like nuclear energy while strengthening transmission networks.

Looking ahead, HTS technologies are expected to play a critical role as key enabling technologies. HTSs, in particular, with their potential to reduce transmission losses and realize high-capacity, high-efficiency power systems, are anticipated to underpin the large-scale integration of renewable energy and enhance system resilience.